

1.4

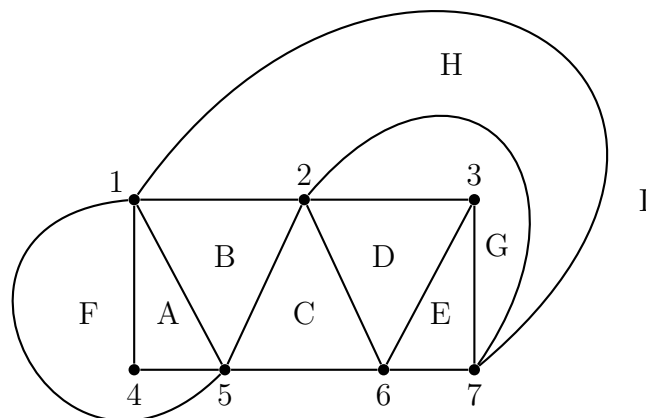
7. In each case, give the values of $\mathbf{r}$, $\mathbf{e}$, or $\mathbf{v}$ (whichever is not given) assuming that the graph is planar. Then either draw a connected, planar graph with the property, if possible, or explain why no such planar graph can exist.

(d) 14 edges and nine regions

$$r = 9, e = 14, v = ?$$

$$\begin{aligned} * r &= e - v + 2 \\ \Rightarrow 9 &= 14 - v + 2 \\ \Rightarrow -7 &= -v \\ \Rightarrow \boxed{v = 7} \end{aligned}$$

$$\begin{aligned} * e &\leq 3v - 6 \\ \Rightarrow 14 &\leq 3(7) - 6 = 15 \checkmark \end{aligned}$$

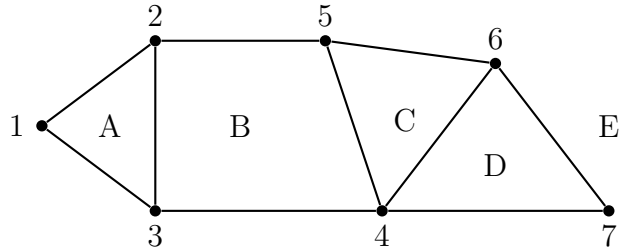


(f) Five regions and 10 edges

$$r = 5, e = 10, v = ?$$

$$\begin{aligned}
 * r &= e - v + 2 \\
 \Rightarrow 5 &= 10 - v + 2 \\
 \Rightarrow -7 &= -v \\
 \Rightarrow \boxed{v = 7}
 \end{aligned}$$

$$\begin{aligned}
 * e &\leq 3v - 6 \\
 \Rightarrow 10 &\leq 3(7) - 6 = 15 \checkmark
 \end{aligned}$$



8. If a connected planar graph with n vertices all of degree 4 has 10 regions, determine n .

$$\begin{aligned}
 * \sum_{i=1}^{i=k} \deg(n_i) &= 2e \\
 \Rightarrow 4n &= 2e \\
 \Rightarrow e &= 2n
 \end{aligned}$$

$$\begin{aligned}
 * r &= e - v + 2 \\
 \Rightarrow 10 &= 2n - n + 2 \\
 \Rightarrow 10 &= n + 2 \\
 \Rightarrow \boxed{n = 8}
 \end{aligned}$$

16. Prove that if a graph G has 11 vertices, then either G or its complement $\overline{G}$ must be nonplanar.

Proof. By contradiction, suppose G and $\overline{G}$ are planar. Consider the graph K_{11} . We know that

$$\begin{aligned}
 G \cup \overline{G} &= K_{11} \\
 \Rightarrow |E(G)| + |E(\overline{G})| &= |E(K_{11})|.
 \end{aligned}$$

The total number of edges in K_{11} is

$$|K_{11}| = \frac{11(10)}{2} = 55 \text{ edges.}$$

However, since G and $\overline{G}$ are planar, they both satisfy the inequality

$$\begin{aligned} e_G &\leq 3v - 6 \quad \text{and} \quad e_{\overline{G}} \leq 3v - 6 \\ \Rightarrow e_G &\leq 3(11) - 6 \quad e_{\overline{G}} \leq 3(11) - 6 \\ \Rightarrow e_G &\leq 27 \quad e_{\overline{G}} \leq 27. \end{aligned}$$

This means the total edges in G and $\overline{G}$ can be at most 27 each. So,

$$e + \bar{e} \leq 27 + 27 = 54$$

$\Rightarrow |G| + |\overline{G}| \neq 55$. Since both G and $\overline{G}$ satisfy planarity, the sum of their edges can never equal the edges in K_{11} , which is impossible. $\therefore$ either G or $\overline{G}$ must be nonplanar. $\square$